

Measuring the Design Decisions of an Open-Source Post-Quantum HSM Chip

ML-KEM-512 NTT and Keccak, from a Python golden model to a SKY130 layout

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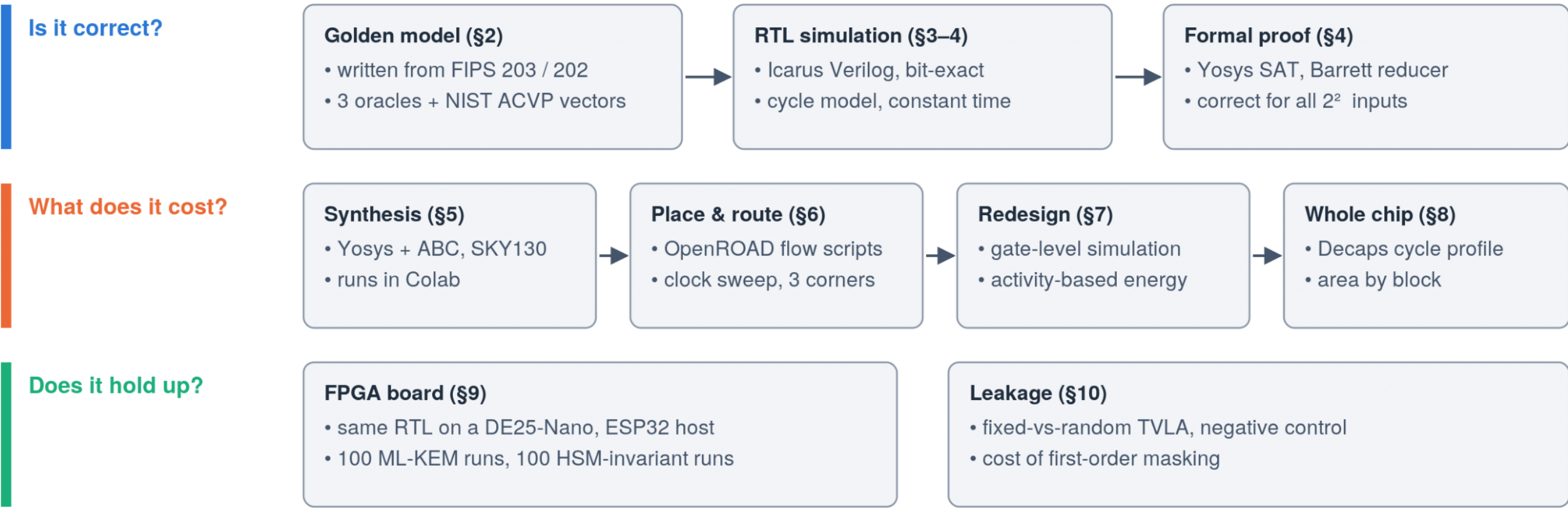


run it in Colab

What did each architectural decision made for the ASIC actually cost in area, latency and energy, what does the design still leak, and can anyone re-derive those numbers with open tools?

How this notebook measures a design decision

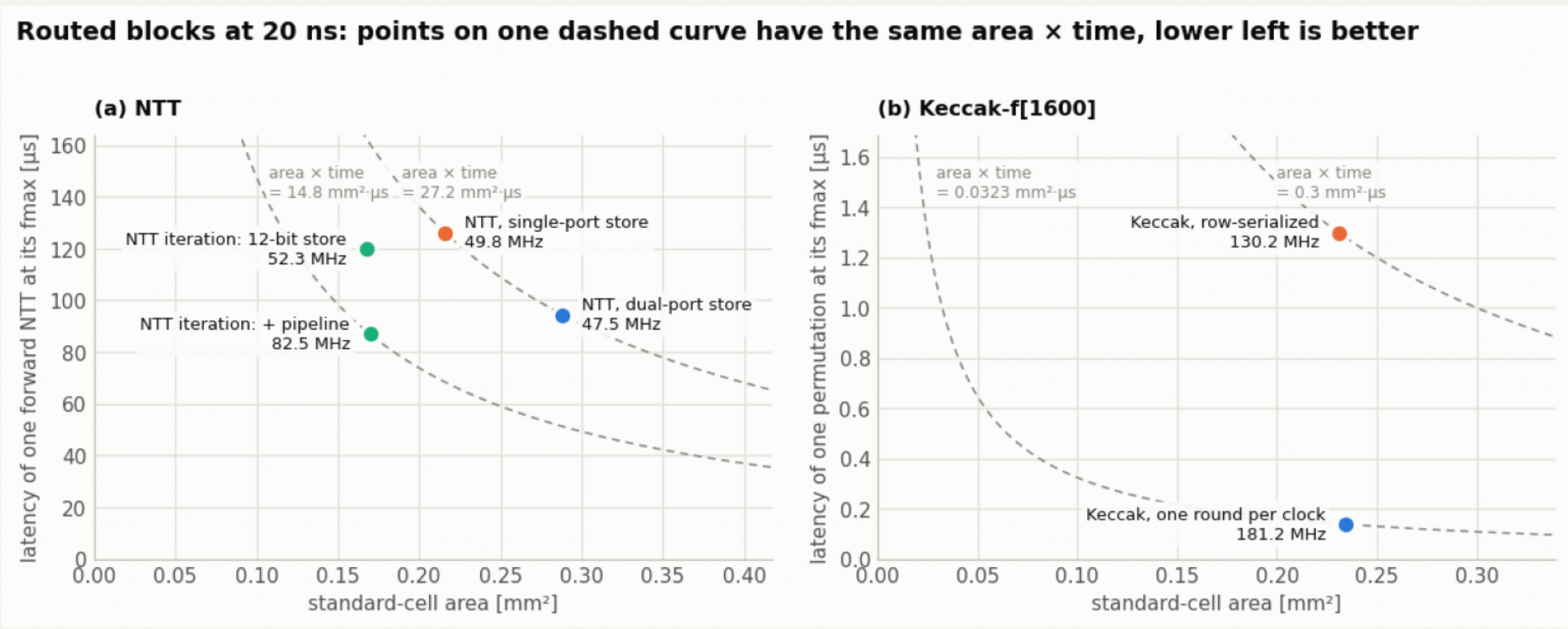
Every step uses open-source tools except the FPGA bitstream; the section number is given in each box.



I. Is it correct?

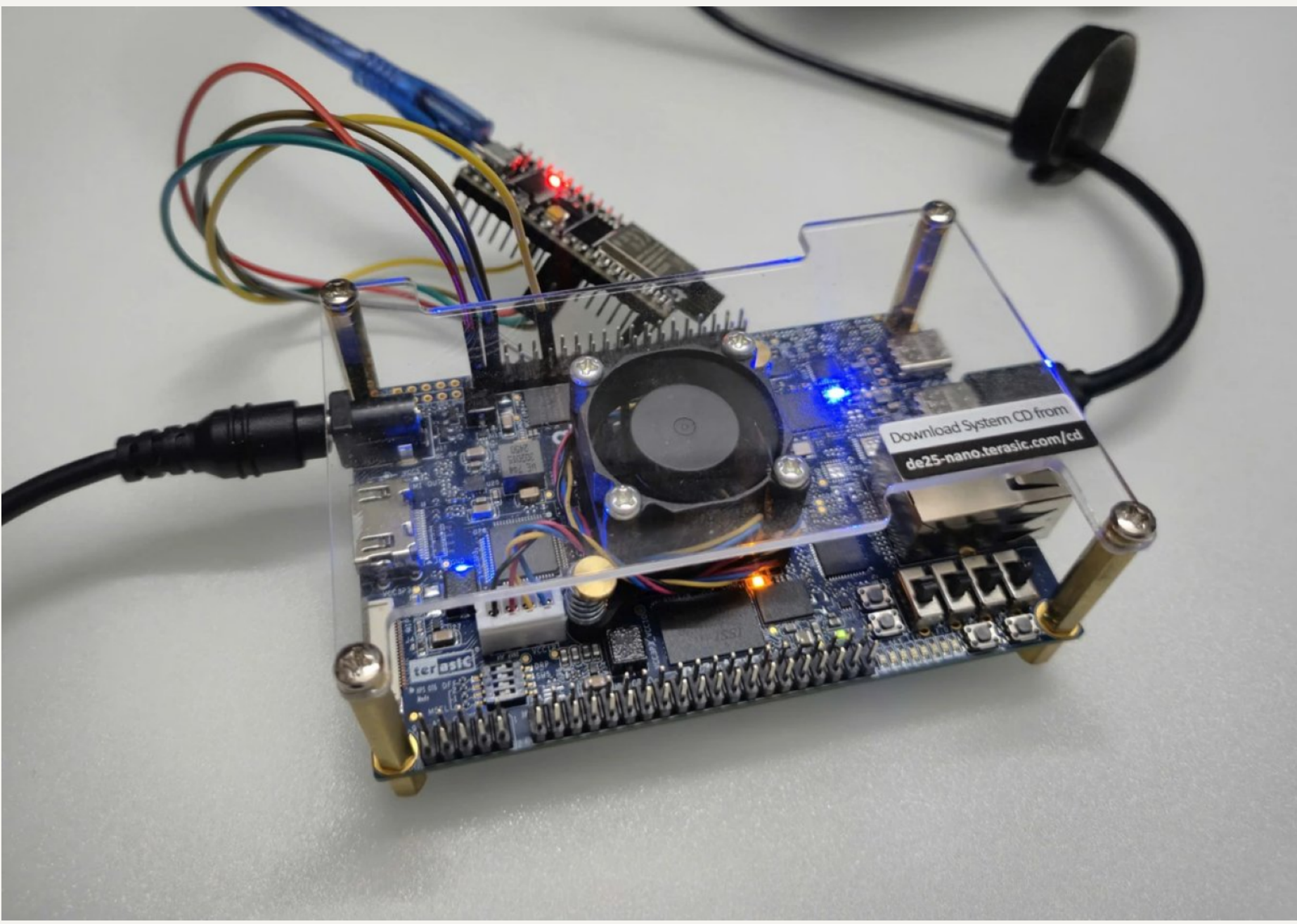
- 25/25 NIST ACVP key generations reproduced byte for byte by the complete RTL
- 0 mismatches over 208,896 NTT coefficients and 404 Keccak permutations, at a constant latency
- The Barrett reducer is proven correct for all 2^{24} inputs, and one subtraction suffices
- The routed netlists reproduce the golden results at gate level

II. What does it cost?

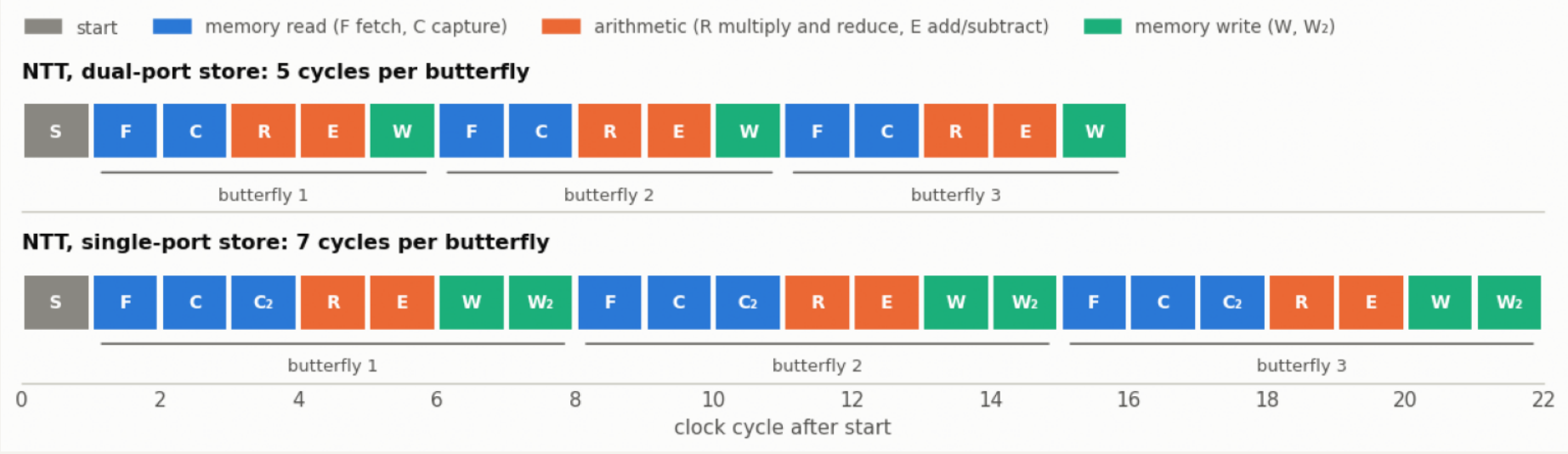


A measured redesign of the NTT, after routing: fmax 50 → 83 MHz, area -21%, area × time -46%.

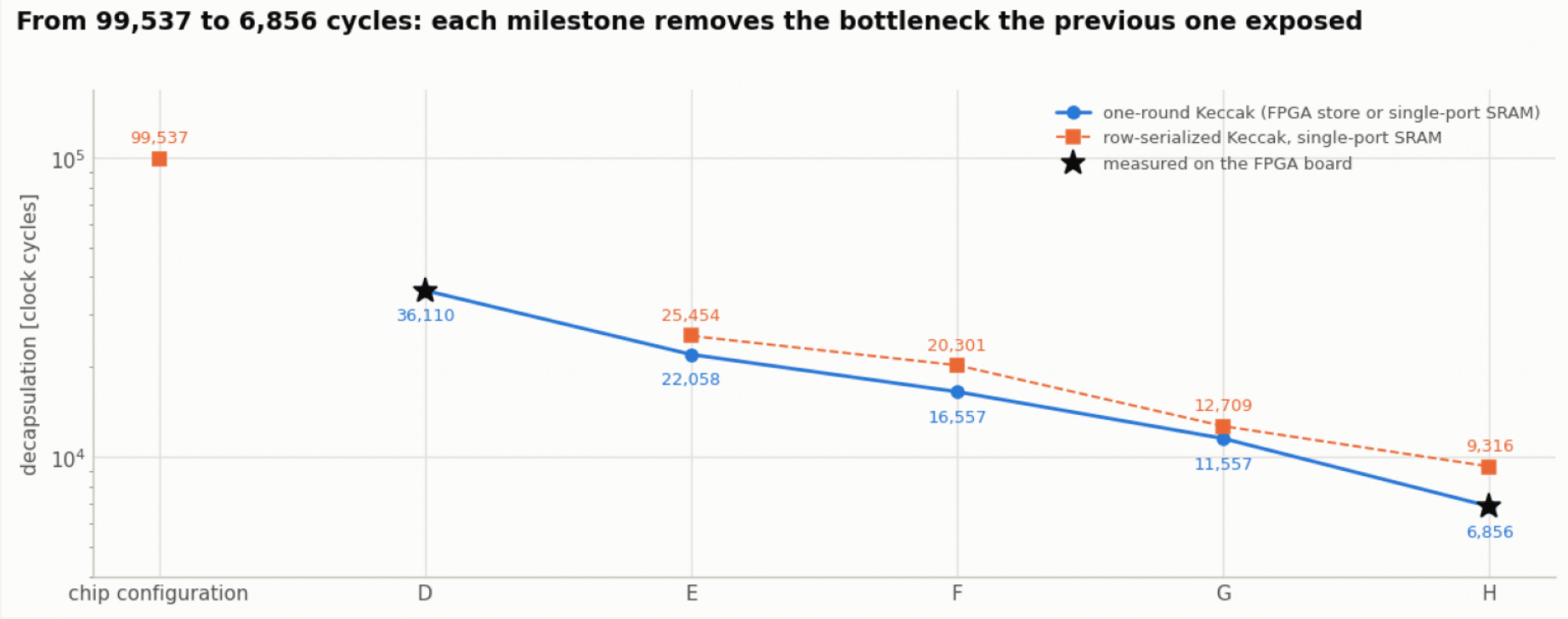
III. Does it hold up?



The same RTL on the DE25-Nano: 100/100 two-role ML-KEM runs pass.

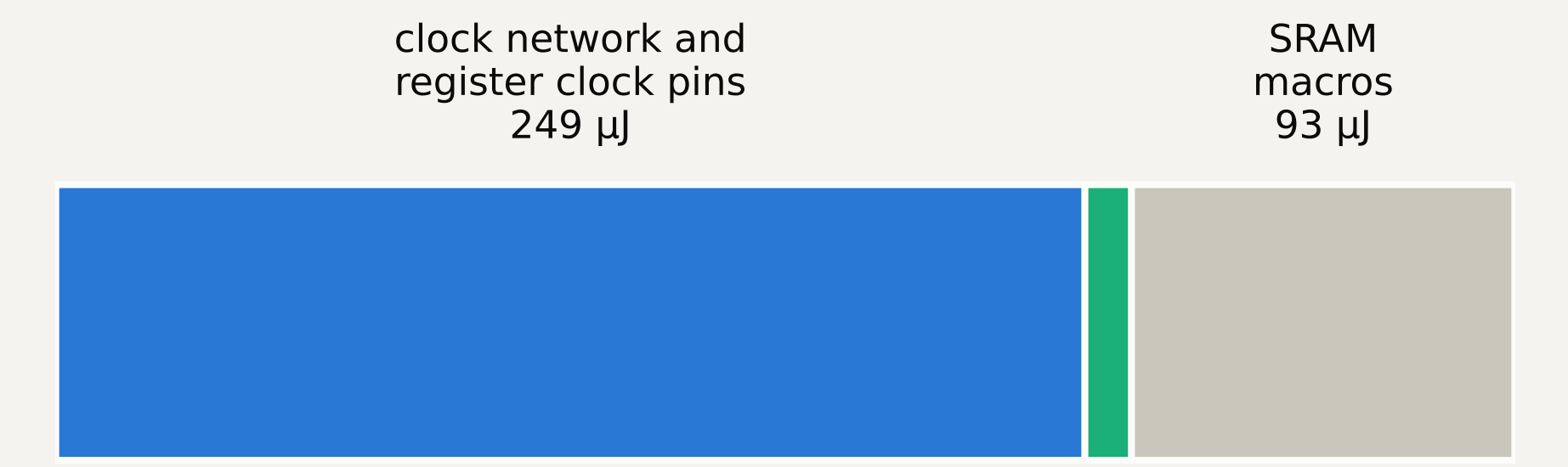
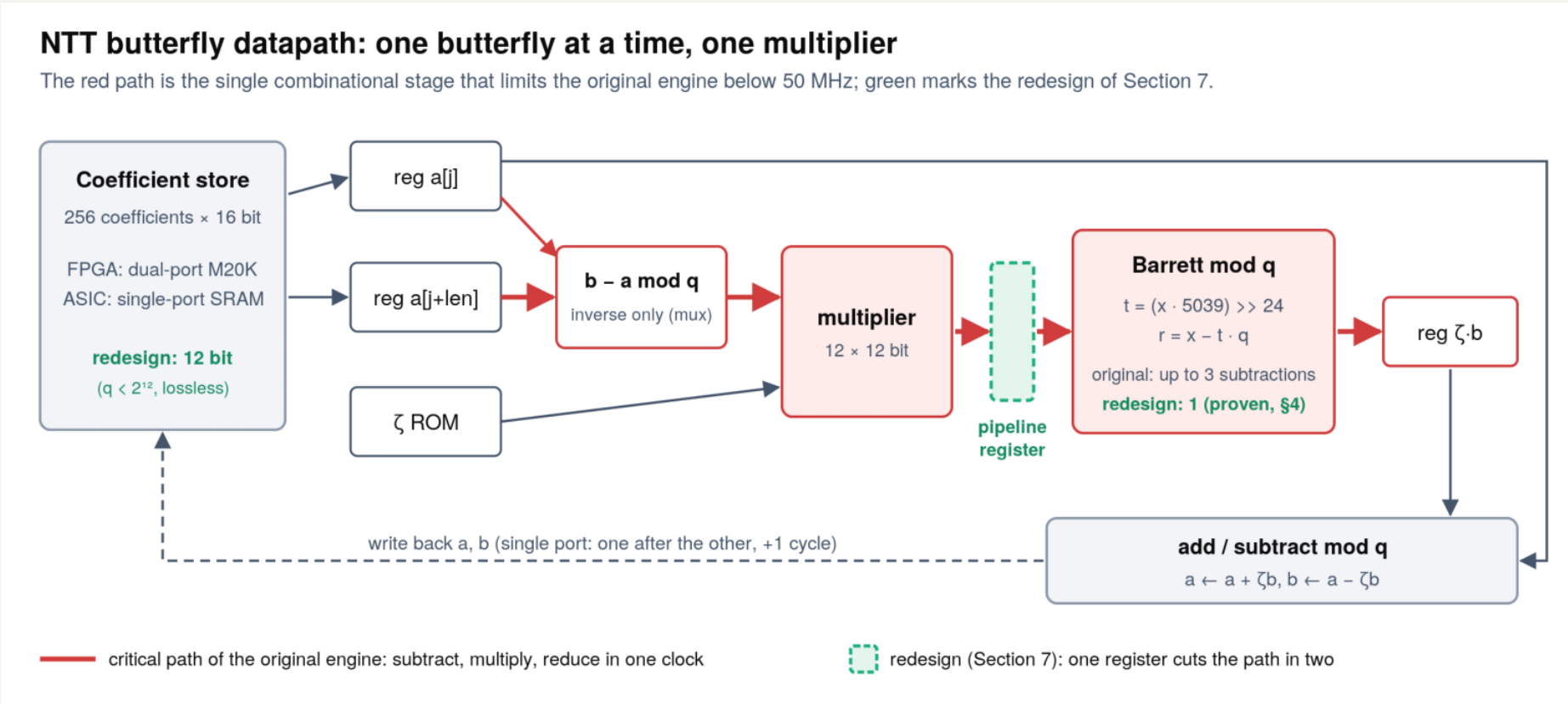


The controller visits the 896 butterflies in the order of FIPS 203, Algorithm 9; the single-port store needs two extra cycles per butterfly.

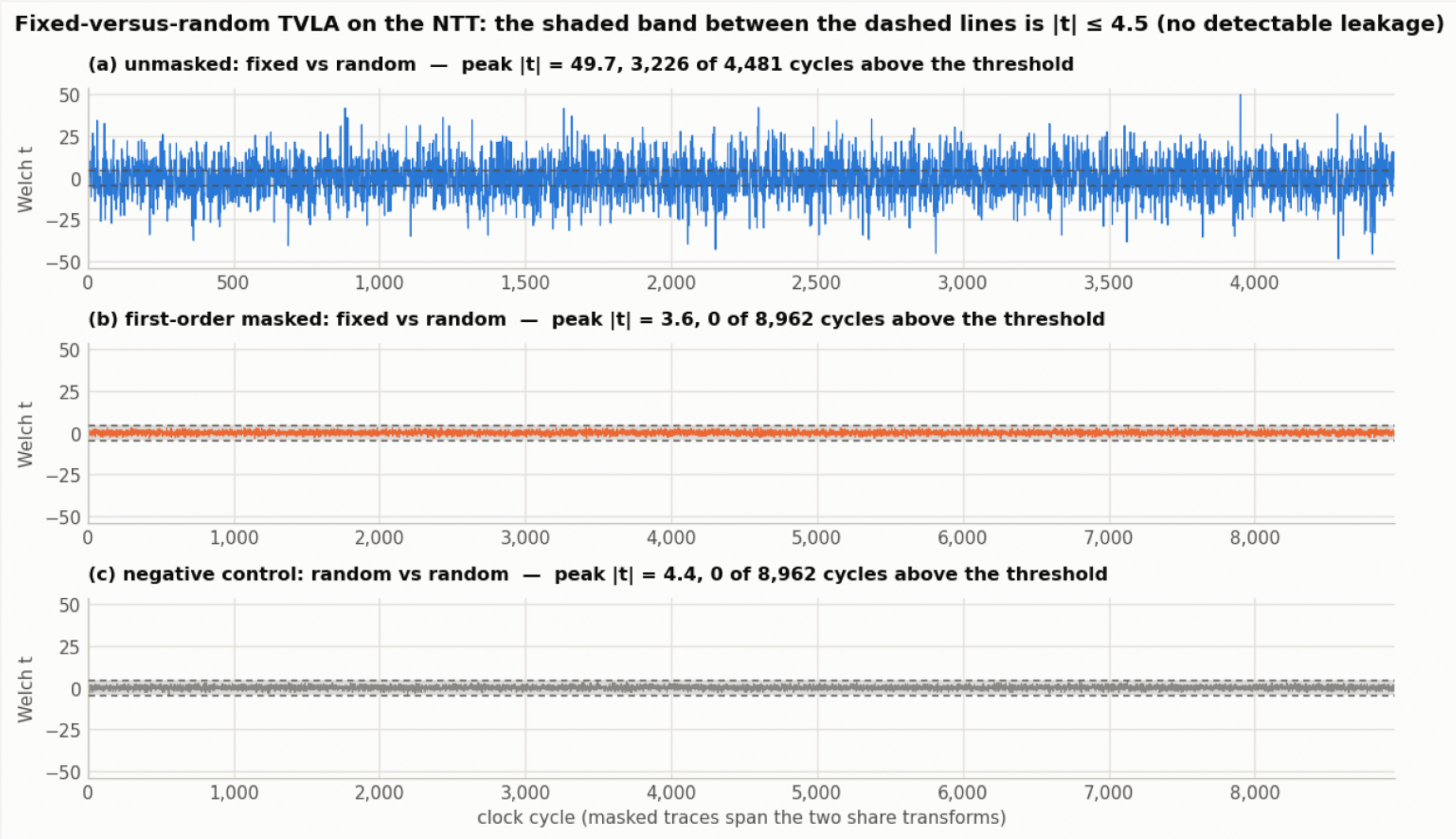


A second redesign stores two coefficients in every SRAM word: a forward NTT takes 14 instead of 130 μ s and 0.16 instead of 0.50 μ J. Streaming data between a two-lane engine and the datapath, with changes that remove no check, a decapsulation takes 6,856 instead of 99,537 cycles, as the board confirms.

One decapsulation on the chip: 0.35 mJ at 25 MHz



The integration, not ML-KEM, sets the energy: gating the clock of idle blocks and the chip selects of the macros would save roughly 40 to 55 %. The NTT is busy 50% of a decapsulation, the Keccak permutation 0.8%.



Simulated TVLA: largest $|t|$ 49.7 unmasked, 3.6 with first-order masking (threshold 4.5).

What transfers

- Weigh a block by how often the system waits for it.
- Measure the integrated chip, not only its blocks.
- Check the check: nine results here first looked right and were not.
- Tie the prose to the data: assertions stop the notebook when they part.